

Autonomous Coal Mine Safety Robot

1. Abstract

Coal mines are hazardous environments due to the presence of toxic gases, poor visibility, and underground obstacles. Traditional monitoring systems rely on fixed sensors and manual inspection, which expose workers to dangerous conditions. This project proposes an AI-based autonomous coal mine monitoring robot using ESP32 as the main controller. The system integrates MQ-4 and MQ-7 gas sensors for methane and carbon monoxide detection, an ultrasonic sensor for obstacle avoidance, and an ESP32-CAM for live video streaming.

The robot operates in both autonomous and WiFi-controlled manual modes. Sensor data is processed using AI-based threshold logic to classify gas levels and trigger emergency actions when necessary. Real-time video and gas data are transmitted to a mobile or PC for remote monitoring. This system reduces human risk and improves safety in underground mining environments.

2. Problem Statement

Traditional coal mine monitoring systems rely on fixed gas sensors and manual inspection, which expose workers to hazardous conditions. These systems lack mobility, autonomous navigation, and intelligent real-time monitoring capabilities.

3. Objectives

- To design an autonomous coal mine monitoring robot.
- To detect methane and carbon monoxide gases using MQ-4 and MQ-7 sensors.
- To implement obstacle detection using ultrasonic sensor (HC-SR04).
- To provide live video streaming using ESP32-CAM.
- To enable WiFi-based manual control.
- To integrate AI-based safety decision logic.

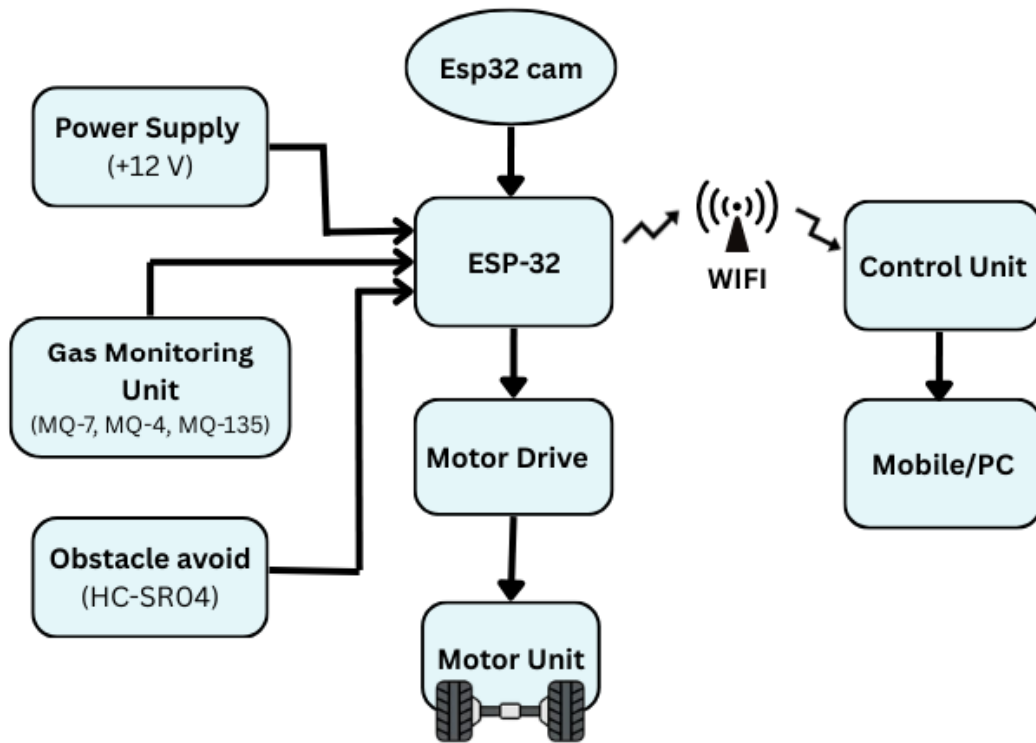
4. Existing System

Current systems include fixed gas detection units, CCTV surveillance, manual inspection, and portable gas detectors. These systems lack mobility, intelligent decision-making, and real-time integrated monitoring.

5. Proposed System

The proposed system is a mobile AI-based robotic platform capable of autonomous navigation, real-time gas detection, live video transmission, WiFi-based remote control, and emergency safety override.

6. Block Diagram



7. Methodology

The system collects gas and obstacle data through sensors. ESP32 processes the data using AI-based threshold logic to classify gas levels into Safe, Warning, and Danger. In autonomous mode, the robot navigates automatically and avoids obstacles. In manual mode, users control the robot through a WiFi-based web interface. If dangerous gas levels are detected, the system overrides manual control and triggers emergency actions.

8. AI Integration

AI is implemented through gas level classification, emergency override mechanisms, and intelligent obstacle avoidance logic. Future enhancements include predictive gas analysis using machine learning.